MIL-T-25439A(USAF)
25 April 1962
Superseding
MIL-T-25439 (USAF)
6 February 1956

MILITARY SPECIFICATION

TRANSMITTER, RATE OF FLOW, IN-FLIGHT REFUELING, 0-10,000 PPM, TYPE MB-3

1. SCOPE

1.1 This specification covers the requirements for one type of in-flight-refueling fuel flow rate transmitter, designated Type MB-3.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-W-76	Wire and Cable, Hook-Up, Electrical, Insulated
MIL-V-173	Varnish, Moisture- and Fungus-Resistant, for the Treatment of Communications, Electronic, and Associated Electrical Equipment
MIL-M-3171	Magnesium Alloy, Processes for Corrosion Protection of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-G-5572	Gasoline, Aviation: Grades 80/87, 91/96, 100/130, 115/145
MIL-J-5624	Jet Fuel, Grades JP-3, JP-4, and JP-5

FSC 6680

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MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-P-8422	Power Supply, 28V DC Input, 15V 4 Cycles, 3 Phase output, Type MC-1
MIL-F-8615	Fuel System Components, General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-T-9107	Test Reports, Preparation of
MIL-I-25440	Indicator, Rate of Flow and Consumption, Type MA-9
MIL-A-25441	Amplifier, Electronic Control, Type MF-1
MIL-M-45202	Magnesium Alloys; Anodic Treatment of
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MS3106	Connector, Plug, Electric, Straight
MS3108	Connector, Plug, Electric, Angle 90°
MS25103	Switches, Toggle, Double Pole, Three-Hole Mounting
MS33586	Metals, Definition of Dissimilar

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 QUALIFICATION.- The transmitter furnished under this specification shall be a product which has been tested and has passed the qualification tests herein and has been listed on or approved for listing on the applicable qualified products list.

3.2 SELECTION OF SPECIFICATIONS AND STANDARDS.- Specifications and standards for necessary commodities and services not specified herein shall be selected according to MIL-STD-143.

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3.3 MATERIALS

3.3.1 NONMAGNETIC MATERIALS.- Nonmagnetic material shall be used for all parts of the system except where magnetic materials are essential.

3.3.2 FUNGUS-PROOF MATERIALS.- Materials that are not nutrients for fungi shall be used to the greatest extent practicable. In cases where materials that are nutrients for fungi must be used, such material shall be protected or treated with a fungicidal agent in accordance with MIL-V-173, or shall be protected by hermetic sealing.

3.3.3 METALS.- Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.3.3.1 DISSIMILAR METALS.- Unless suitably protected against electrolytic corrosion by means of protective coatings or hermetic sealing, intimate contact of dissimilar metals shall be avoided. Dissimilar metals are defined by MS33586.

3.3.4 CHOICE OF MATERIALS.- Materials shall conform to applicable specifications as specified herein. When materials are used which are not specifically designated, they shall be entirely suitable for the purpose. The use of lightweight materials and weight-saving designs is a major consideration, and their use shall be investigated and exploited to the greatest possible extent.

3.3.5 CORROSIVE FUMES.- Materials used in the indicator construction shall not produce corrosive or deleterious fumes under the test conditions specified herein.

3.3.6 PROTECTIVE TREATMENT.- When materials are used in the construction of the transmitter that are subject to corrosion due to salt air or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification.

3.4 DESIGN AND CONSTRUCTION.- The transmitter shall be designed to provide an electrical output signal for a fuel flow range of 0 to 10,000 ppm. The signal shall be suitable for remote indication and for use in operating an indicator in accordance with MIL-I-25440 with an amplifier in accordance with MIL-A-25441. The design shall be based on a principle that is not fundamentally sensitive to density changes.

3.4.1 OUTPUT SIGNAL.- The electrical signal shall be single-phase, 400-cycle a-c voltage. The magnitude of the voltage shall be 1V per 1,000 ppm fuel flow under the standard test conditions. The electrical output signal shall have a carrier frequency identical to the input

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power frequency. The output voltage signal shall be at a leading phase angle of 15° with respect to the input voltage.

3.4.2 EXCESS FLOW.- The transmitter shall not be damaged by fuel flow rates in excess of the rated value provided the pressure rating of the transmitter is not exceeded.

3.4.3 FUEL VARIATIONS.- The transmitter shall be designed to operate satisfactorily when using fuel having any of the variations in standard characteristics specified in MIL-F-5624 and MIL-F-5572.

3.4.4 ADJUSTMENT AND REPAIR.- The transmitter shall be so designed that adjustment and repairs can be easily made by personnel of operating units and overhaul bases.

3.4.5 The transmitter shall be so constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.4.5.1 Pivots, bearings, and gears shall not bind and shall be as nearly frictionless as practicable.

3.4.5.2 Screws and nuts shall be safetied with lockwire or suitable lockwashers or locknuts where necessary to meet the vibration requirements and other conditions incident to shipping, installation, and service.

3.4.6 OPERATING PRESSURE.- The transmitter shall be designed for a normal operating pressure of 120 psi, a proof pressure of 240 psi, and a burst pressure of 480 psi.

3.5 PART NUMBERING OF INTERCHANGEABLE PARTS.- All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification part number requirements of MIL-D-70327 shall govern the manufacturer's part numbers and changes thereto.

3.6 PERFORMANCE.- The transmitter shall be capable of satisfactory operation when subjected to the following conditions:

a. Temperature - Temperatures ranging from -54° to +71° C

b. Humidity - Relative humidity up to 100 percent

c. Salt spray - Exposure to salt sea atmosphere

d. Vibration - Vibration incident to service use

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e. Fungus - Fungus growth as encountered in tropical climates

f. Explosion proof - Operation in an explosive vapor within or surrounding the equipment

g. Altitude - Pressures ranging from 30 inches Hg down to 3.4 inches Hg (approximately an altitude of 50,000 feet)

h. Shock - Shock forces as anticipated

i. Dielectric strength - A potential of 550V, 60 cyc es ac for a period of 10 seconds when applied as specified in 4.6.

j. Leakage - A hydraulic proof pressure of 240 psi applied and maintained for 5 minutes

k. Cycling operation - The system shall be capable of operating for 500 hours of actual or simulated flow through the transmitter without failure as specified in 4.6.21.

3.6.1 RADIO NOISE INTERFERENCE.- The transmitter shall meet the radio noise interference requirements of MIL-I-6181, as modified herein.

3.6.2 CONTAMINATED FUEL OPERATION.- The transmitter shall operate satisfactorily with fuels contaminated to the extent specified in 4.6.23.

3.7 ELECTRICAL POWER AND WIRING

3.7.1 The electrical power for the transmitter motor shall be obtained from a Type MC-1 power supply conforming to MIL-P-8422.

3.7.2 The electrical power for the output signal circuit shall be 115 ±5V, 400 ±20 cps. There shall be no voltage phase angle difference between this power and the power to the rest of the system comprised of the indicator and amplifier. The wave form of the supply voltage may be nonsinusoidal with a crest factor of 1.414 ±10 percent and with harmonics having respective amplitudes that are less than 5 percent of the fundamental.

3.7.3 Hookup wire and external unshielded connector cable shall be in accordance with MIL-W-76.

3.7.4 When using the terminating impedance specified in the standard test conditions, the accuracy of the output signal shall not be adversely affected by a connecting cable up to a total length of 200 feet.

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3.7.5 INTERNAL WIRING.- Where practicable, the internal wiring shall be color coded.

3.7.6 ELECTRICAL PARTS.- All standard parts, such as resistors, condensers, and transformers, shall conform to MIL-E-5400.

3.7.7 EXTERNAL WIRING.- The in-flight refueling system interconnecting wiring diagram shall conform to figure 1.

3.8 DENSITY CORRECTION.- The transmitter shall measure the fuel flow rate on a gravimetric basis without a separate and distinct density-correcting means.

3.9 FUEL LINE CONNECTIONS.- The fuel line connections shall be as specified by the procuring activity and shall be furnished with suitable plugs or caps to prevent access of foreign material during storage and shipment.

3.9.1 MOUNTING.- The transmitter shall be designed for mounting by means of the fuel line connections or by straps.

3.9.2 LINE SIZE.- The fuel system tubing adjacent to the transmitter shall be in accordance with the requirements of the procuring activity.

3.10 DAMPING.- The transmitter shall be suitably damped.

3.11 DIMENSIONS.- The outline dimensions of the transmitter shall conform to figure 2.

3.12 FLOW DIRECTION.- The direction of fuel flow through the transmitter shall be permanently and legibly marked on each side by an arrow as shown on figure 2.

3.13 WEIGHT.- The weight of the transmitter shall not exceed 18.0 pounds.

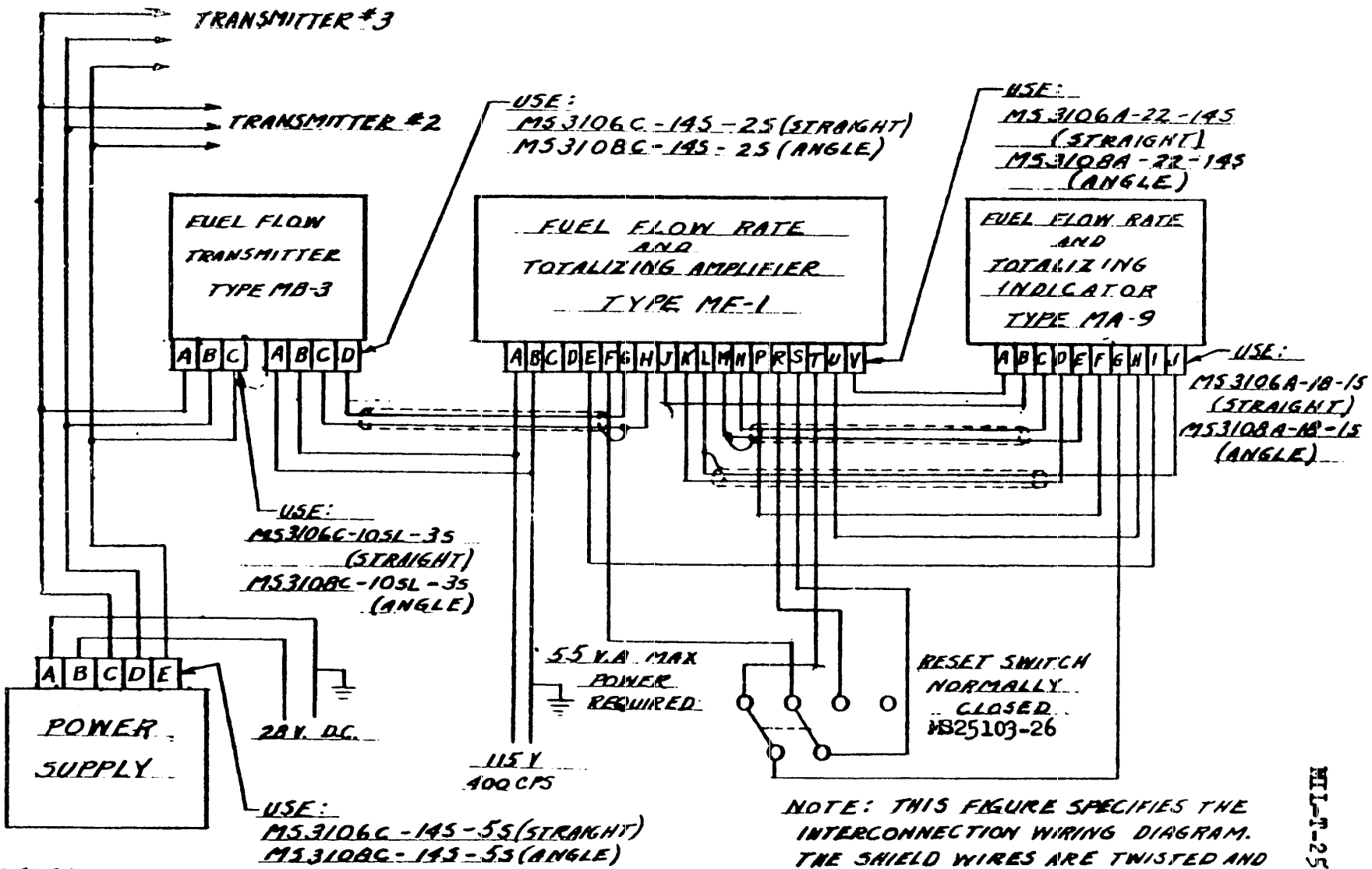
3.14 SCREW THREADS.- Unless otherwise specified, the threads of all machine screws 0.060 inch or larger in diameter shall conform to MIL-S-7742.

3.15 FINISHES AND PROTECTIVE COATINGS

3.15.1 Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625 except as follows:

3.15.1.1 Aluminum alloys that do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

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NOTE: THIS FIGURE SPECIFIES THE INTERCONNECTION WIRING DIAGRAM. THE SHIELD WIRES ARE TWISTED AND THE SHIELD IS GROUNDED AT THE AMPLIFIER TO REDUCE HUM PICK UP.

NOTES:
1. VOLTAGE DROP FROM 115 VOLT SOURCE TO PINS "A" AND "B" OF TRANSMITTER AND AMPLIFIER MUST BE EQUAL.
2. WIRES INSIDE SHIELDS SHALL BE TWISTED.
3. SHIELDS SHALL BE SOLDERED TO CONNECTOR AT AMPLIFIER ONLY.

FIGURE 1 INFLIGHT REFUELING SYSTEM-INTERCONNECTING WIRING.

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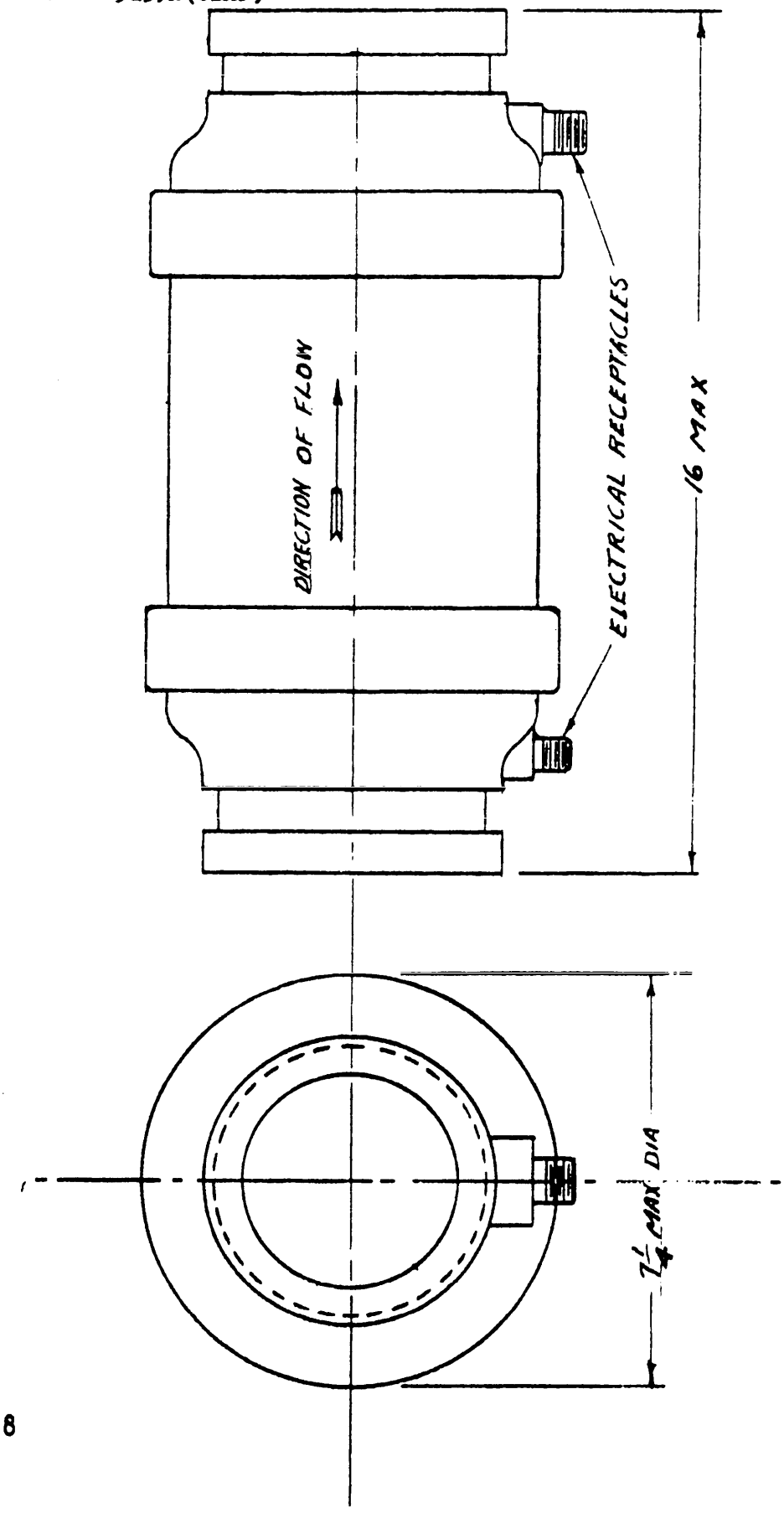


FIGURE 2 TRANSMITTER

DIMENSIONS IN INCHES

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3.15.1.2 Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatment in accordance with MIL-C-5541 may be used in lieu of anodizing.

3.15.1.3 Castings containing nonaluminum alloy inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

3.15.1.4 Small holes and inserts and small areas that must be machined after the part is anodized need not be anodized.

3.15.1.5 When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.15.2 Magnesium alloy parts shall be covered with an HAE type anodic film conforming to MIL-M-45202, type II, class A, grade 3.

3.15.2.1 When abrasion resistance is a factor, chemical films in accordance with MIL-M-3171 shall not be used in lieu of the anodic-type film specified in 3.15.2.

3.15.3 Steel parts shall be cadmium plated, where practicable, in accordance with QQ-P-416, type II or III, and of a class that is adequate to achieve the degree of protection required.

3.15.4 Finishes and protective coatings that will crack, chip, or scale during normal service life or due to extremes of atmospheric conditions shall not be used.

3.16 IDENTIFICATION OF PRODUCT.- Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.17 WORKMANSHIP.- The transmitter, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding, brazing, painting, riveting, machine screw assemblies, and freedom of parts from burrs and sharp edges.

3.17.1 DIMENSIONS AND TOLERANCES.- Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the transmitter or the system, they shall be held or limited accordingly.

3.17.2 SCREW ASSEMBLIES.- Assembly screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw or bolt or threads.

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3.17.3 RIVETING.- Riveting operations shall be carefully performed to be sure that the rivets are tight and satisfactorily headed.

3.17.4 GEARS.- Gear assemblies shall be properly aligned and meshed, and shall be operable without interference, tight spots, loose spots, or other irregularities.

3.17.5 CLEANING.- All parts shall be thoroughly cleaned and ·ose, spattered, or excess solder, metal chips, and other foreign material removed after final assembly. Burrs and sharp edges as well as resin flash that may crumble shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 SUPPLIER'S RESPONSIBILITY.- The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 CLASSIFICATION OF TESTS.- The inspection and testing of transmitters shall be classified as follows:

a. Qualification tests

b. Acceptance tests.

4.3 TEST CONDITIONS

4.3.1 STANDARD ATMOSPHERIC CONDITIONS.- Whenever the pressure and temperature are not specified, it is understood that the test will be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately +25° C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 TAPPING.- Except where otherwise specified, the transmitter shall be gently tapped or vibrated before a test reading is taken. An electrical vibrator set at 60 cps may be used. If a vibrator is used, the maximum double amplitude shall be 0.002 inch when attached to the test item.

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4.3.3 TEST POSITION.- Unless otherwise specified, the transmitter shall be tested in the normal operating position. The transmitter will be in the normal operating position when horizontal with the electrical connectors at the 3 o'clock position when viewed from the inlet end of the transmitter.

4.3.4 TEST POWER.- Unless otherwise specified, the test power to the transmitter shall be as follows:

4.3.4.1 The 3-phase motor power into the transmitter motor shall be 4 cycles. This frequency shall be held by the power supply to within ±0.1 percent. The a-c voltage fed into the motor power supply inverter shall be 28.0 ±0.5V dc.

4.3.4.2 The voltage input to the output signal network shall be 115V ac ±0.1 percent and a frequency of 400 ±2 cycles.

4.3.4.3 The 400-cycle supply voltage may be nonsinusoidal with a crest factor of 1.414 ±10 percent and with harmonics whose respective amplitudes are less than 5 percent of the fundamental.

4.3.5 TEST FUEL.- Unless otherwise specified, the fuel used in testing or for calibration purposes shall conform to MIL-F-5624 (JP-4). The specific gravity of the fuel shall be 0.785 ±0.005 at room temperature. (Note: A commercial equivalent with higher flash point may be used.)

4.3.6 MASTER FLOWMETER.- A master fuel flowmeter to establish flow rate may be used. The calibration of this master shall be verified by the procuring activity.

4.3.7 MASTER VOLTMETER.- The output signal from the transmitter shall be measured with a master voltmeter specially designed for the purpose and accurate to within ±0.1 percent. The input impedance of the voltmeter shall be 100,000 ohms ±5 percent resistive load. The voltmeter shall be calibrated in rms voltage. The voltmeter reading shall be proportional to the peak value of the a-c voltage. The calibration of the voltmeter shall be established by comparison with an electrodynamometer-type voltmeter calibrated with a pure sine wave. As an alternate, the master voltmeter may consist of a transducer mulled against the output signal.

4.3.8 PHASE ANGLE METER.- The phase angle meter shall be specially designed for the purpose and shall be accurate to ±0.1°. (See 6.3.2.)

4.4 QUALIFICATION TESTING

4.4.1 TEST SAMPLES.- The qualification test samples shall consist of three transmitters representative of the production equipment. The

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samples shall be identified by the manufacturer's part number and such other information as required by the procuring activity.

4.4.2 TEST REPORT.- When tests are conducted at a location other than the laboratory of the procuring activity, three copies of a test report in accordance with MIL-T-9107 shall be furnished to that activity.

4.4.3 QUALIFICATION TESTS.- The qualification tests shall consist of all tests described under TEST METHODS.

4.5 ACCEPTANCE TESTS.- Acceptance tests shall consist of:

a. Individual tests

b. Sampling plans and tests.

4.5.1 INDIVIDUAL TESTS.- Each transmitter shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Dielectric strength

c. Leak test

d. Scale error and friction

e. Phase angle tolerance.

4.5.2 SAMPLING PLANS AND TESTS

4.5.2.1 SAMPLING PLAN A.- Two transmitters shall be selected at random from each 100 or less produced on the contract or order and subjected to the following tests:

a. Position error

b. Low temperature scale error and friction

c. Low temperature leak

d. High temperature scale error and friction

e. Vibration resonance

f. Vibration cycling.

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4.5.2.2 SAMPLING PLAN B.- Unless otherwise specified, two transmitters shall be selected at random from the first 10 transmitters on the contract or order and subjected to the following tests. It is permissible to run these tests on as many as four groups in parallel to reduce the elapsed time for completing the tests. Where more than one group is subjected to these tests, each test shall be run on a minimum of two transmitters and all interrelated tests shall be run on the same transmitters.

a. Individual tests

b. Sampling plan A tests

c. Voltage and frequency variation

d. Power consumption

e. Electrical characteristics

f. Radio noise interference

g. Pressure drop

h. Response time

i. Substitute fuel

j. Fuel resistance test

k. Humidity

l. Fungus

m. Salt spray

n. Altitude

o. Explosion proof

p. Shock test

q. Cycling operation

r. Reverse flow test

s. Contaminated fuel test.

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4.5.2.3 REJECTION AND RETEST.- When an item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After corrections have been made, all tests shall be repeated.

4.5.2.3.1 INDIVIDUAL TESTS MAY CONTINUE.- For production reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or produced later shall not be made until it is determined that all items meet all the requirements of the specification.

4.5.2.3.2 DEFECTS IN ITEMS ALREADY ACCEPTED.- The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 TEST METHODS

4.6.1 EXAMINATION OF PRODUCT.- Each transmitter shall be visually examined externally to determine conformance to this specification with respect to nameplates, markings, workmanship, and finish.

4.6.2 DIELECTRIC STRENGTH.- All pins of the transmitter electrical receptacles, except those which are grounded or used for bonding, shall be shorted together and a potential of 550V, 60 cycles ac applied between the shorted pins and the receptacle for a period of 10 seconds. No insulation breakdown shall result.

4.6.3 LEAK TEST.- The transmitter shall be connected to a hydraulic pump or other source of hydraulic pressure and a proof pressure of 240 psi applied and maintained for a period of 5 minutes. There shall be no leakage or damage as a result of this test.

4.6.4 SCALE ERROR AND FRICTION.- The test points shall be as noted in table I. The scale error shall not exceed the tolerance in table I. The friction error shall be determined at the mid-scale point by bringing the flow up to, but not exceeding, the specified rate and then down to, but not below, the specified rate. The difference between the upscale and downscale readings shall not exceed the tolerance in table II. Tapping shall not be used when making the friction error readings.

4.6.4.1 The transmitter shall be tested on a flow stand capable of determining the true flow within an accuracy of ±0.5 percent of the point. The reference standard for verification of this accuracy shall be the flow stand facility at ASD.

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TABLE I

Scale Error Tolerance ~~±~~PPM

Flow PPM	Room Temp	Low Temp	High Temp	Subst Fuel	Cycling Test
0	20	30	30	30	40
1,000	20	30	30	30	40
2,000	25	35	35	35	45
3,000	40	55	55	55	65
4,000	60	80	80	80	90
6,000	70	100	100	100	110
8,000	120	180	180	180	190
10,000	220	290	290	290	300

TABLE II

Test	Tolerance
Friction -	
Room temperature	5 ppm
Low temperature	5 ppm
High temperature	5 ppm
Position error	20 ppm
Voltage and frequency effect	
Scale error	20 ppm
Phase angle	3°
Vibration	
Variation	40 ppm
Oscillation	60 ppm (double amp)

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4.6.5 PHASE ANGLE TOLERANCES.- The phase angle tolerances at the test points shall not exceed the values shown in table III.

TABLE III

Flow PPM	Phase Angle Tolerance, Degree ±		
	Room Temp	Low Temp	High Temp
0	--	--	--
2,000	1.0	1.5	1.5
4,000	0.5	1.0	1.0
6,000	0.5	1.0	1.0
8,000	1.0	1.5	1.5
10,000	1.5	2.0	2.0

4.6.6 POSITION ERROR.- In determining the position error, two additional calibration tests shall be conducted. The first test position shall be the normal test position. The second test position shall be horizontal with the connectors at the 6 o'clock position when looking at the inlet end of the transmitter. The third test position shall be horizontal with the connectors at the 9 o'clock position when looking at the transmitter from the inlet end. The calibration errors in positions 2 and 3 shall not exceed the room temperature tolerance in position 1 by more than that specified in table II.

4.6.7 LOW TEMPERATURE SCALE ERROR AND FRICTION.- The transmitter shall be subjected to a temperature of -55° ±2° C for a period of 4 hours. While still at this temperature, the transmitter shall be tested for scale error and friction as specified in 4.6.4. The scale error and the friction error shall not exceed the tolerances specified in tables I, II, and III. The fuel used in this test shall be at room temperature, and the ambient temperature around the transmitter shall be maintained at -55° ±2° C.

4.6.8 LOW TEMPERATURE LEAK.- The transmitter shall be subjected to a temperature of -55° ±2° C for a period of 4 hours. While the transmitter and fuel are still at this temperature, the test specified in 4.6.3 shall be repeated. There shall be no leaks and the transmitter shall not be damaged as a result of this test. This test may be combined with the test specified in 4.6.7.

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4.6.9 HIGH TEMPERATURE SCALE ERROR AND FRICTION.- The transmitter shall be subjected to a temperature of 70° ±2° C for a period of 4 hours. While still at this temperature, the transmitter shall be tested for scale error and friction error as specified in 4.6.4. The scale error and friction error shall not exceed the tolerances in tables I, II, and III. The fuel used in this test shall be at room temperature, and the ambient temperature around the transmitter shall be maintained at 70° ±2° C.

4.6.10 VIBRATION RESONANCES.- For the vibration resonances tests, the transmitter shall be subjected to the following frequency range and vibratory accelerations or double amplitudes:

Frequency (cpm)	Amplitude
300 - 6,000	0.010 in.
6,000 - 30,000	±5g
30,000 - 60,000	±5g

4.6.10.1 When conducting the vibration resonance test and observing errors, the fuel flow rate shall be approximately 1,000 ppm. During this test, when the output signal error is not being observed, the vibration may be applied with zero flow but with fuel in the transmitter.

4.6.10.2 The transmitter shall be electrically connected to a master voltmeter. The resonant frequency of the transmitter shall be determined by varying the frequency of applied vibration slowly through the range of frequencies and vibratory acceleration specified. The resonant frequency shall be determined by observing the output signal on the master voltmeter. The output signal vibration variation error and oscillation error shall be noted during the scanning of the frequency range for the resonant frequency. These vibration errors shall not exceed the tolerance specified in table II.

4.6.10.3 The test specimen shall be vibrated at the indicated resonant condition for 1 hour (time specified refers to one axis of vibration). During this period, the applied double amplitude or vibratory acceleration shall be as specified in 4.6.11.1. The periods of vibration shall be accomplished in sequence for each of three mutually perpendicular axes of vibration applied along any one axis. Each test period shall be conducted at the most severe resonance frequency.

4.6.10.4 When resonant frequencies are not apparent within the specified frequency range, the transmitter shall be vibrated twice as long as specified for resonance at a frequency of 3,000 cpm. At the end of this test, the transmitter shall be inspected to determine that no damage or defects have resulted due to this test. The transmitter shall then be subjected to the vibration cycling test specified in 4.6.11.

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4.6.11 VIBRATION CYCLING.- The transmitter shall be mounted and connected as specified in 4.6.10.1 and 4.6.10.2 and vibrated with the frequency cycling between 600 and 60,000 cpm in 15-minute cycles at the applied amplitude specified in 4.6.11.1. The vibration cycling shall be conducted for a period of 1 hour in each of 3 mutually perpendicular axes. After the vibration test, a scale error test shall be conducted and the readings established shall not exceed the room temperature tolerances specified in table I. The test specimen shall then be thoroughly examined, and there shall be no damage as a result of this test.

4.6.11.1 Vibration conditions during cycling shall be in accordance with the following:

Frequency (cpm)	Amplitude
300 - 600	0.050 in.
600 - 4,500	0.036 in.
4,500 - 30,000	±10g
30,000 - 60,000	±5g

4.6.12 VOLTAGE AND FREQUENCY VARIATION EFFECT.- The applied voltage for this test shall be set at the following points:

a. 22-1/2V dc
110V ac
380 cps

b. 24V dc
110V ac
420 cps

c. 24V dc
120V ac
380 cps

d. 30V dc
120V ac
420 cps

4.6.12.1 When the applied voltages and frequencies are changed as specified, the errors in the output signal shall not exceed the room temperature tolerances specified in tables I and III by more than the tolerance in table II. (Note: Proper allowance shall be made for change of a-c voltage from standard condition.)

4.6.13 POWER CONSUMPTION.- With nominal supply voltage applied to the transmitter, the power consumption shall not exceed the following maximum values:

4.6.13.1 The motor power shall not exceed 30 watts which shall be determined by measuring the d-c voltage input to the master power supply and the d-c amperes into the inverter of the master power supply. The watts shall be the product of volts and amperes.

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4.6.13.2 The power consumed by the output signal circuit shall not exceed 7 va when measured at the nominal test voltage.

4.6.14 ELECTRICAL CHARACTERISTICS.- Sufficient electrical tests shall be performed by the manufacturer on the transmitter and its components to insure satisfactory performance when connected in a complete flowmeter system as specified in 3.4.

4.6.15 RADIO NOISE.- The radio noise test shall be in accordance with MIL-I-6181, except that the frequency range of 0.15 to 150 megacycles shall apply for radiated radio noise.

4.6.16 PRESSURE DROP.- The pressure drop of the transmitter shall be determined for the flow rates specified in table IV and shall not exceed the values specified therein.

TABLE IV

Flow PPM	Max Pressure Drop PSI
2,000	0.25
4,000	1.0
6,000	2.3
8,000	4.0
10,000	6.25

4.6.16.1 The pressure-drop values specified shall not be exceeded whether the transmitter is functioning or not functioning.

4.6.17 RESPONSE TIME.- The fuel flow shall be changed from 0 to 8,000 ppm by means of a rapid-action valve in the fuel line. The time for the indication of a master voltmeter connected to the transmitter to read within 1 percent of the steady state position shall not exceed 1 second.

4.6.18 SUBSTITUTE FUEL TEST.- The transmitter shall be tested for output signal error with fuels in which the densities are approximately 10 percent above and 10 percent below, respectively, the calibrating fuel density used. The output signal errors shall not exceed the values in table I.

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4.6.19 FUEL RESISTANCE TEST.- The transmitter shall be tested for fuel resistance in accordance with MIL-F-8615.

4.6.20 ENVIRONMENTAL TESTS.- The following environmental tests shall be conducted in accordance with the specified procedures of MIL-E-5272 and as specified herein.

4.6.20.1 HUMIDITY.- The humidity test shall be conducted in accordance with procedure I. Immediately following the completion of the fifteenth cycle, the transmitter shall be subjected to the test specified in 4.6.4.1. The scale errors and friction errors shall not exceed the tolerances specified in table I.

4.6.20.2 FUNGUS RESISTANCE.- The fungus test shall be conducted in accordance with procedure I. This test shall not be required on hermetically sealed units.

4.6.20.3 SALT SPRAY.- The salt spray test shall be conducted in accordance with procedure I. At the completion of this test, the transmitter shall be subjected to the tests specified in 4.6.3 and 4.6.4. There shall be no leakage and the scale errors shall not exceed the tolerances specified in table I.

4.6.20.4 ALTITUDE.- The altitude test shall be conducted in accordance with procedure VI, condition C, for a period of 4 hours for electrical components which are not hermetically sealed. Fuel flow will not be required during this test.

4.6.20.5 EXPLOSION PROOF.- The explosion proof test shall be conducted in accordance with procedure III.

4.6.20.6 SHOCK.- The shock test shall be in accordance with procedure IV. The transmitter shall be subjected to impact shocks of 15g, each shock impulse having a time duration of 7 ±2 milliseconds. Three shocks shall be applied in each of the following directions:

a. Along the longitudinal axis

b. At 45° to the longitudinal axis in one direction.

At the conclusion of this test, there shall be no evidence of damage, and the transmitter shall meet the room temperature tolerances.

4.6.21 CYCLING OPERATION.- The system shall be operated continuously for 500 hours with actual or simulated flow through the transmitter. The actual or simulated cycling rate from zero to full scale and back to zero shall be once each 2 ±1 minute. No failure of any components shall occur.

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4.6.21.1 The calibration of the transmitter shall be recorded before and after the 500-hour cycling test. The tolerances shall not exceed the values specified in table I.

4.6.21.2 SIMULATED FLOW.- The following simulated flow procedure may be used in lieu of 500 hours of cycling with actual flow:

a. The transmitter shall be operated for 50 hours with random flow variations.

b. The transmitter shall be disassembled by separating the turbine end from the impeller end.

c. The impeller shall be operated for 400 hours by exciting the drive motor in the normal manner. The impeller shall be immersed in fuel in the normal manner during this operation.

d. The turbine end shall be operated for 400 hours by applying a mechanical torque to the turbine causing it to deflect to full scale and back to zero at a rate of once every 2 ±1 minute. The turbine shall be immersed in fuel in the normal manner during this operation, and the 115V 400-cycle power excitation shall be applied in the normal manner.

e. The transmitter shall be reassembled and operated for 50 hours with random flow variations.

Random flow variations of the type that are obtained in a flow stand when calibrating production transmitters are suitable for the operation specified in a and e.

4.6.22 REVERSE-FLOW TEST.- The transmitter shall be subjected to a rapidly applied reverse flow. This flow shall build up from 0 to a value at least 60 percent of full scale in approximately 1 second. After five surges of reverse flow, the transmitter shall be retested in the normal manner and the scale error shall not exceed the room temperature tolerance specified in table I.

4.6.23 CONTAMINATED FUEL OPERATION.- The transmitter shall be operated with contaminated fuel (80 grams of foreign matter per 1,000 gallons of fuel) at a flow rate of approximately 2,000 ppm for 1 hour. It shall then be operated with normal test fuel and a scale and friction error test conducted immediately. It shall operate satisfactorily and be within room temperature tolerances. The foreign matter shall consist of not less than 68 percent SiO_2 and shall have a particle size analysis as shown in table V.

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TABLE V

Foreign Particle Size, Microns	Percent of Total Weight
0 - 5	39 ± 2
5 - 10	18 ± 3
10 - 20	16 ± 3
20 - 20	18 ± 3
over 40	9 ± 3
Through a 200 mesh screen	100

5. PREPARATION FOR DELIVERY

5.1 PACKAGING AND PACKING.- Preservation, packaging, and packing of the transmitter shall conform to MIL-P-7936.

5.2 MARKING OF SHIPMENTS.- Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

Transmitter, Rate of Flow, In-Flight
Refueling, 0-10,000 PPM, Type MB-3

5.2.1 REINSPECTION MARKINGS.- Reinspection date markings shall be in accordance with the instructions of the procuring activity.

6. NOTES

6.1 INTENDED USE.- The Type MB-3 transmitter covered by this specification is intended for use in a fuel flowmeter system used for aircraft in-flight-refueling operations.

6.2 ORDERING DATA.- Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Information, if any, to be submitted with qualification samples.

c. Whether or not sampling plan B tests will be omitted.

d. Level of packaging and packing desired.

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6.3 DEFINITIONS

6.3.1 HERMETIC SEAL.- A hermetic seal is defined as a perfectly closed and airtight seal made between vitric or metallic materials, or both. A hermetic seal is not intended to include seals accomplished by gaskets or "O" rings.

6.3.2 PHASE ANGLE.- The phase angle is the electrical angle of input voltage and output signal voltage measured at the zero voltage point on the sine wave as the voltage goes from negative to positive.

6.4 QUALIFICATION.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Aeronautical Systems Division, Attn: ASNPF, Wright-Patterson Air Force Base, Ohio, and information pertaining to qualification of products may be obtained from that activity.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any right or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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